

GEETHANJALI V N

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Education

Velammal Engineering College

September 2023 – May 2027

B.E. Computer Science and Engineering — CGPA: 7.87/10

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Technical Skills

Languages: Java, Python, SQL, JavaScript

Web & Backend: HTML, CSS, FastAPI, Flask, REST APIs

Databases: MySQL, MongoDB, DynamoDB, PostgreSQL, Redis

Cloud & DevOps: AWS (Lambda, API Gateway, EC2, S3, IAM), Docker, Git, GitHub, Agile, Scrum

Core CS: OOP, Operating Systems, Computer Networks, DBMS, System Design

Competitive programming : LeetCode | HackerRank | Skillrack

Projects

Research Paper Agent | *LangGraph, Gemini, ChromaDB, FastAPI, Sentence Transformers*

June 2026

- Architected a multi-step LangGraph agentic pipeline that autonomously retrieves, ranks, and synthesizes 20+ academic papers per query from arXiv and Semantic Scholar, cutting manual literature review time by 70%.
- Implemented RAG pipeline with ChromaDB vector store and sentence-transformer embeddings; grounded all model responses in cited sources, eliminating hallucinated references.
- Automated literature review generation, methodology comparison, and research-gap identification — producing structured outputs (summaries, presentation outlines, insights) in under 60 seconds.
- Built REST API layer using FastAPI to serve agent queries with asynchronous execution, supporting concurrent research sessions.

DocuMind AI | *FastAPI, Gemini, OCR, PostgreSQL, AWS S3*

April 2026

- Developed an AI document intelligence platform processing 5+ document types (invoices, receipts, scanned PDFs) using Gemini multimodal LLM + Tesseract OCR, achieving 90%+ structured field extraction accuracy.
- Designed confidence-score-based human-review routing that reduced manual verification workload by 60% by auto-approving high-confidence extractions above a configurable threshold.
- Built REST APIs with FastAPI; persisted structured extraction results and raw documents to PostgreSQL and AWS S3, enabling full audit trail and re-processing capability.
- Implemented multimodal LLM pipeline converting unstructured PDFs and images to validated JSON records with schema enforcement and error recovery.

URL Shortener | *FastAPI, Redis, MySQL, Docker*

December 2025

- Engineered a high-throughput URL shortening service supporting custom aliases and handling 10k+ daily redirects, containerized via Docker for portable deployment.
- Achieved 40% reduction in p99 lookup latency by implementing a Redis TTL-based caching layer on top of MySQL, eliminating redundant database round-trips on hot URLs.
- Developed RESTful APIs for URL creation, retrieval, expiry management, and analytics tracking with per-URL click metrics and geographic data.

Experience

TAMIZHAN SKILLS

June 2025 – July 2025

IoT & Embedded Systems Intern

Remote

- Built REST API endpoints to stream live sensor telemetry to a web dashboard, enabling real-time occupancy visualization across 20 parking slots.
- Diagnosed and resolved I2C/UART communication faults between microcontroller and sensor modules, improving data-packet success rate from 70% to 98%.

Achievements & Certifications

- **Won First Prize – Blockathon'24, Velammal Engineering College** Aug 2024
- **Top 105 out of 10,000+ global submissions — Google On Campus Solution Challenge (Top 1%)** May 2025
- **NPTEL Deep Learning (Elite Certification) – IIT Ropar** Apr 2026
- **HackerRank Software Engineer Intern Certification** Jun 2026
- **Artificial Intelligence Primer Certification – Infosys Springboard** May 2026